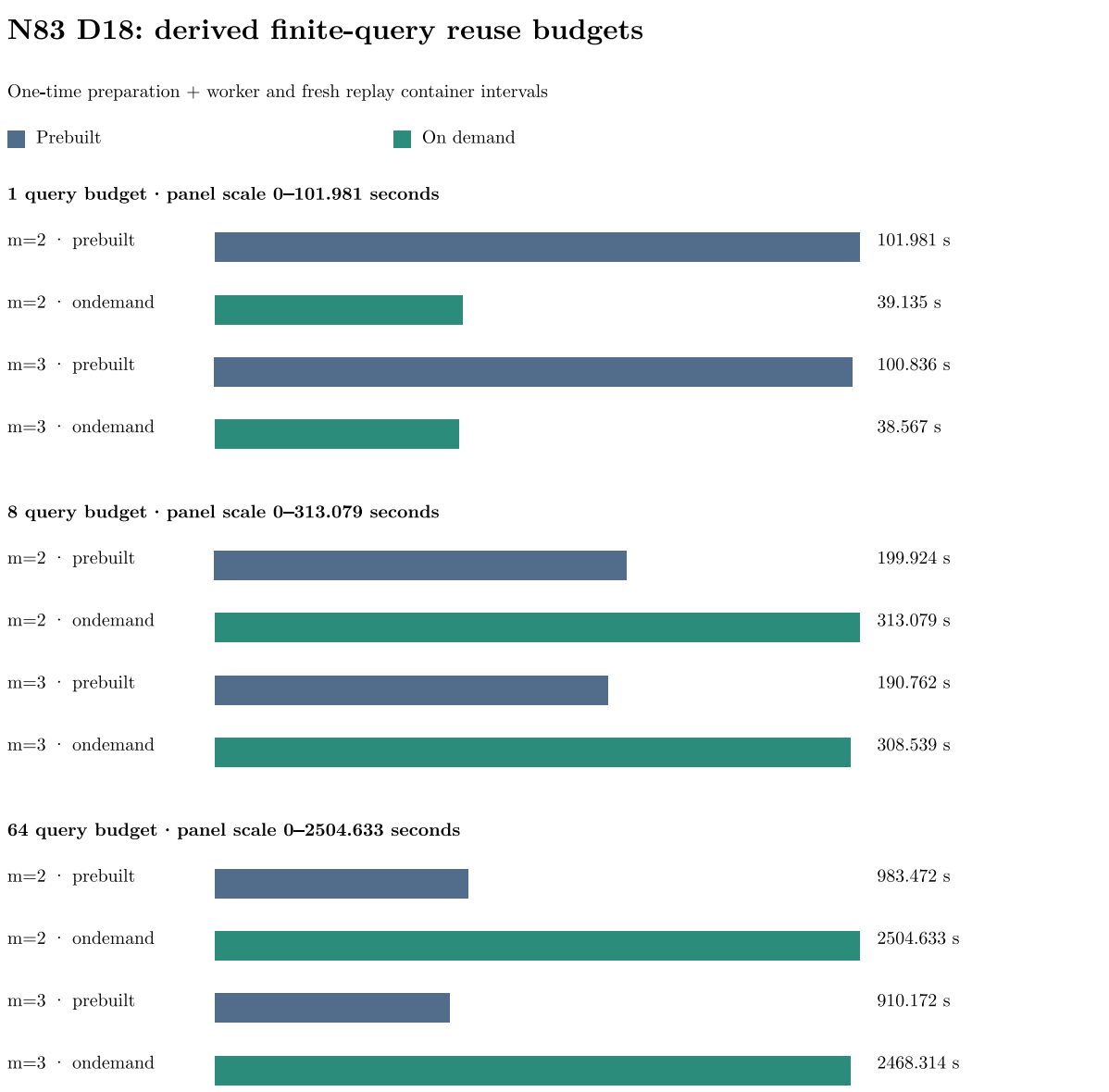
N83 finite-query reuse accounting

2026-10-20 · accounting addition from retained native measurements

The frozen preparation protocol requests reuse counts 1, 8 and 64. The earlier report supplied 1 and 8. This native re-derivation completes all three and separately charges fresh per-query receipt replay and the worker container boundary. It introduces no new timed query or algorithm.



Panel scales differ and are printed above each group. Every value is a projection from nine complete probes per mode and m, on eight public points with target 0 repeated. The 64-query panel was not executed. Prebuilt adds 87.989088451 s for census/export/full-object replay; on-demand preparation occurs inside each worker interval.

Accounting addition, not an algorithm improvement. All source observations are shared-host L0. Docker launch/teardown outside the containers, remote custody/storage, earlier retry history, later rank/linear algebra/individual logarithm work and matched rho are excluded. Source: accounting.json; native derivation: examples/n83_prebuild_reuse_accounting.rs.

Accounting correction and scope

The earlier sums used census + export + generic whole-base replay + the mean worker interval for each projected query. They omitted the fresh per-query receipt replay and the worker interval outside its internal timer. Those earlier sums remain a separately labeled view in `accounting.json`; the figure includes both observed container intervals. It still is not a complete cold IC budget.

Native reconstruction

The Rust tool verifies the pinned manifest and all 833 retained files in `paired-mode-v1`. It requires all 36 expected probes, their exact finite attempt counts and identical D point/label sets, all 36 bound fresh replay receipts, successful non-OOM container exits, and nanosecond start/end timestamps on the pinned run date.

Every cell has nine probe observations: two at public target 0 and one at each of indices 1 through 7. The arity-2 event scans all 2,760,082 distinct first summands; the arity-3 event draws 2^20 pairs. The retained finite events have zero relation rows and rank zero. These counts do not estimate unrestricted natural yield.

m	Mode	Worker mean (s)	Worker container (s)	Fresh replay container (s)
2	prebuilt	9.848484	9.900365	4.091555
2	ondemand	35.130566	35.182691	3.952197
3	prebuilt	8.828438	8.880963	3.965646
3	ondemand	34.601494	34.651515	3.915884

Prebuilt setup is 87.989088451 s, the sum of the separately pinned census, export and generic full-object replay. Those one-time phases and the query worker have distinct recorded native revisions. They ran on the same shared L0 SSH host. The point and coefficient label sets agree between query preparation modes.

For each mode and m, the new projection is setup + q × (mean worker container interval + mean fresh replay container interval). The older worker-only projection is setup + q × mean worker interval. On-demand enumeration/closure/indexing are inside every on-demand worker, so its separate setup term is zero. It is not free preparation.

Evidence and unresolved charges

The native regression test rejects failed/capped container outcomes, nonpositive intervals, wrong dates and malformed fractional timestamps. The accounting run completed with exit zero, after checking the immutable input digests. CI repeats the native computation and requires byte-identical JSON. The graph and table read that JSON directly.

Worker peak RSS reached 651,149,312 bytes in the prebuilt cells and 563,486,720 bytes on demand. This is observed worker RSS, not a measured peak for an unexecuted 64-query panel. All per-arm records are retained.

Excluded charges remain explicit: Docker launch/teardown outside the container intervals; upload, custody and storage; earlier implementation and infrastructure retry history; natural rank collection outside these finite events; linear algebra and individual logarithm verification; same-point rho and an external-fastest comparison. The card's original 16 min 48 s host interval includes its setup, controls, repeats and launch/replay operations and is not replaced by a projection.

Candidate, selected total runtime, complete IC seconds and matched-rho gap remain null. This is an accounting completion, with no measured algorithm improvement or qualified timing promotion.

Sources and reproduction

`examples/n83_prebuild_reuse_accounting.rs`; `PAIRED_MODE_PROTOCOL.md`; `runs/paired-mode-v1.sha256`; the census/export/replay receipts cited by `accounting.json`; `native-test.log`, `native-run.log` and `native-run.exit` beside this report. The original result/graph remain under `reports/result-20261019/`. This addendum and its SVG/150 dpi PNG/PDF have matching browsing copies under `benchmarks/20261020-n83-prebuild-reuse/`.